

A VALENTINE-SHAPED CARDIAC SILHOUETTE IN FELINE THORACIC RADIOGRAPHS IS PRIMARILY DUE TO LEFT ATRIAL ENLARGEMENT

TRISHA J. OURA, AISHA N. YOUNG, BRUCE W. KEENE, IAN D. ROBERTSON,
DENNIS E. JENNINGS, DONALD E. THRALL

Conflicting information has been published regarding the cause of a valentine-shaped cardiac silhouette in dorsoventral or ventrodorsal thoracic radiographs in cats. The purpose of this retrospective, cross-sectional study was to test the hypothesis that the valentine shape is primarily due to left atrial enlargement. Images for cats with a radiographic valentine-shaped cardiac silhouette and full echocardiography examination were retrieved and independently reviewed. A subjective scoring system was used to record severity of radiographic valentine shape. Subjective radiographic evidence of left atrial enlargement in a radiographic lateral projection and a final diagnosis based on medical records were also recorded. A total of 81 cats met inclusion criteria. There was a strong positive correlation ($P < 0.001$) between echocardiographic left atrial size and severity of radiographic valentine shape. There was no effect of echocardiographic right atrial size on the severity of valentine shape, except when concurrent with severe left atrial enlargement. In this situation, right atrial enlargement increased the likelihood of observing a severe valentine shape. There was no effect of right atrial enlargement on the shape of the cardiac silhouette when left atrial enlargement was absent or only mild to moderate. There was no correlation between the category of final diagnosis of cardiac disease and the severity of valentine shape. Findings from this study supported the hypothesis that a valentine-shaped cardiac silhouette in radiographs is due primarily to left atrial enlargement in cats, with right atrial enlargement only impacting the shape if concurrent with severe left atrial enlargement. © 2014 American College of Veterinary Radiology.

Key words: atrium, cat, cardiomegaly, radiography, valentine shaped.

Introduction

THORACIC RADIOGRAPHS ARE STANDARD DIAGNOSTIC tests used in the evaluation of cats with suspected cardiac disease. A large cardiac silhouette can be caused by enlargement of a cardiac chamber, pericardial effusion and peritoneopericardial hernias, great vessel enlargement at the heart base, heart base masses, myocardial masses, and abundant adjacent fat.¹ Cardiac enlargement on thoracic radiographs is used as an important criterion for differential diagnoses, further diagnostic testing recommendations, and disease prognosis. Disease monitoring is also commonly

performed through sequential thoracic radiographic evaluation. Currently, there is conflicting information regarding the cause of a “valentine-shaped” cardiac silhouette on dorsoventral (DV) or ventrodorsal (VD) feline thoracic radiographs. A valentine shape refers to a somewhat triangular cardiac silhouette with a wide base and narrow apex.¹ This pattern of enlargement has been proposed to be due to right atrial displacement, biatrial enlargement,¹ or severe left atrial enlargement^{2,3} but a consensus has not been reached.

In hypertrophic cardiomyopathy (HCM), left atrial enlargement is a common echocardiographic finding that is seen concurrently with a valentine-shaped cardiac silhouette on radiographs.² Cats with HCM are more likely to have left atrial enlargement than biatrial enlargement^{4,5} supporting the hypothesis that a valentine-shaped cardiac silhouette is caused mainly by left atrial enlargement. However, when echocardiographic reports were compared with radiographs of 35 cats with hypertrophic, restrictive, or unclassified cardiomyopathy and a valentine-shaped cardiac silhouette in a previous study, this hypothesis was not supported robustly.³ Specifically, in 15 cats the left atrium was the only dilated chamber, while 13 had biatrial enlargement,

From the Department of Molecular Biomedical Sciences, North Carolina State University College of Veterinary Medicine, Raleigh, N.C. (Oura, Robertson, Thrall); Department of Companion Animals, Atlantic Veterinary College, University of Prince Edward Island, Prince Edward Island, Canada Clinical Sciences, Ross University School of Veterinary Medicine, (Young); Department of Clinical Sciences, North Carolina State University College of Veterinary Medicine, Raleigh, N.C. (Keene); Independent Consultant, Libertyville, IL (Jennings), and IDEXX Telemedicine Consultants (Thrall). Dr. Oura's current address is Clinical Sciences, Cummings School of Veterinary Medicine at Tufts University.

Address correspondence and reprint requests to Trisha J. Oura. Cummings School of Veterinary Medicine, Tufts University. E-mail: trisha.oura@tufts.edu

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suggesting that the specificity of a valentine-shaped cardiac silhouette for diagnosis of left atrial versus biatrial enlargement was low. Also, cats with HCM tended to have left atrial enlargement, while cats with restrictive and unclassified cardiomyopathy tended to have biatrial enlargement.³ Conclusions from this previous study were limited because each cardiomyopathy group included a maximum of 10 cats and the echocardiographic images were not rereviewed to obtain a qualitative judgment of right atrial size.

The aim of the current retrospective, cross-sectional study was to describe associations between a valentine-shaped cardiac silhouette on feline thoracic radiographs and specific chamber enlargement observed in echocardiographic images for a larger sample population of cats. We hypothesized that a valentine-shaped cardiac silhouette on feline thoracic radiographs would be due to left atrial enlargement alone, rather than biatrial enlargement.

Methods

The imaging database at the North Carolina State University Randall B. Terry, Jr. Companion Animal Veterinary Medical Center was searched for all cats that underwent both thoracic radiographic and echocardiographic examinations between the period of January 1, 2004 and July 30, 2012. The radiographic and echocardiographic examinations had to have been performed within seven days of each other in order for cats to be included. The radiology reports for all cats during this time were filtered using the search terms *left atrium*, *left atrial*, *right atrium*, *right atrial*, *biatrial*, and *valentine*, either in the radiographic diagnosis, radiographic findings, or special key words that were recorded by the radiologist at the time the report was generated. *Cardiomegaly* was not used as a search term as it was considered too nonspecific. All radiographs meeting these search criteria were reviewed by a board-certified veterinary radiologist who was unaware of the echocardiographic findings at the time (DT) and were included in the study if at least one lateral view was available and if this observer detected subjective evidence of a valentine appearance of the cardiac silhouette on the VD or DV view.

Whether a DV or VD view was used for analyses depended mainly on whether the study had been ordered by the cardiology (DV) or a noncardiology (VD) service. In patients with both DV and VD radiographs, the DV was used for analyses. If left and right lateral radiographs were available, the best-positioned lateral view was used for analyses.

Either digital or analog images were included in the study. All digital images were acquired with a solid-state, amorphous silicon, indirect flat panel detector (CXDI-50G, Canon Medical Systems, Irvine, California). All analog images were acquired using medium speed, rare earth, screen-film radiography. Analog radiographs had been scanned

previously into DICOM files and were reviewed as digital images. All included echocardiograms were originally performed by a board-certified cardiologist or a directly supervised cardiology resident. All radiographs were retrieved from a PACS and evaluated using an image analysis workstation and designated software (eFilm Workstation, Merge Healthcare, Mississauga, Ontario). All echocardiograms were reviewed offline using a multimodality dynamic image review system (syngo Dynamics workstation, Siemens Medical Solutions, Malvern, Pennsylvania).

A board-certified veterinary radiologist (T.O.) and a radiology intern (A.Y.) reviewed radiographs independently without knowledge of echocardiographic findings. The degree that the cardiac silhouette resembled a valentine shape on DV or VD radiographs was classified using a subjective scoring system. Cats with only slight enlargement of the cardiac base contributing to a triangular cardiac silhouette were graded mild (1), cats with more obvious enlargement of the cardiac base contributing to a triangular cardiac silhouette were graded moderate (2), and cats with marked cardiac base enlargement that occupied a large percentage of the total thoracic cavity width were graded severe (3) (Fig. 1). There was no score for normal cardiac silhouettes as the presence of some degree of a valentine shape was one of the selection criteria for inclusion in the study. Each reader was allowed to assign subjective severity scores based on their own personal experience. On lateral radiographs, left atrial enlargement was evaluated based on the presence or absence of straightening of the caudodorsal margin of the cardiac silhouette and was classified as absent (0) or present (1) (Fig. 2). Following independent assessment by the two observers, consensus agreement was reached between the two observers for all evaluations by reviewing the criteria described above.

A board-certified veterinary cardiologist (B.K.) reviewed echocardiograms without knowledge of thoracic radiographic findings. Still images and cine loops of standard right and left sided parasternal long axis and short axis views, as well as left sided apical 4-chambered views were used for subjective scoring. Left atrial enlargement was scored as absent (1), mild to moderate (2), and severe (3). The size of the left auricle was not independently assessed. Right atrial enlargement was similarly scored as absent (1), mild to moderate (2), and severe (3). In addition, a cardiac diagnosis was recorded for each cat based on results of the echocardiogram, blood pressure, T4 values, and other clinical findings where appropriate.

All statistical tests and analyses were selected and performed by one author (D.J.). A Chi-square test for independence was performed on two-factor tables in Excel (Microsoft Corporation, Redmond, Washington) to test associations between valentine shape severity and diagnoses as well as left atrial enlargement based on echocardiography and consensus score of presence or absence of left

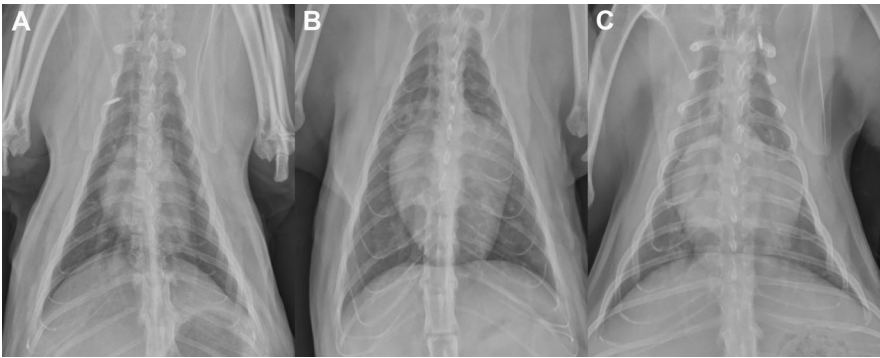


FIG. 1. Representative dorsoventral feline thoracic radiographs illustrating mild (A), moderate (B), and severe (C) valentine shape of the cardiac silhouette as subjectively assessed by a board-certified veterinary radiologist and a veterinary radiology intern.

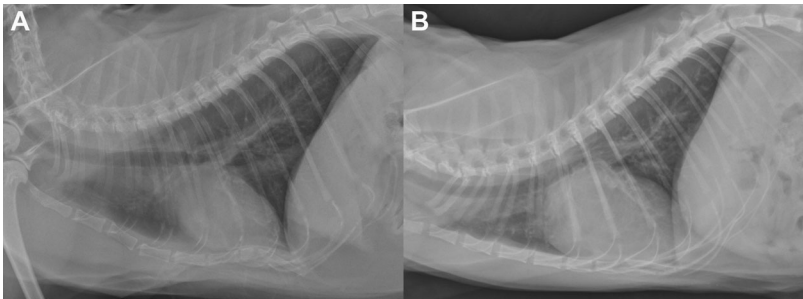


FIG. 2. Representative left lateral (A) and right lateral (B) feline thoracic radiographs illustrating the absence (A) or presence (B) of left atrial enlargement as subjectively assessed by a board-certified veterinary radiologist and a veterinary radiology intern.

atrial enlargement based on thoracic radiographs. A three dimensional table (left atrial grade on echocardiogram, right atrial grade on echocardiogram, degree of valentine shape on DV/VD radiographs) was created, and log linear analysis was performed using the glm function in R (Version 3.0.0 The R Foundation for Statistical Computing Platform, Vienna, Austria). Effects were tested and a model was selected using the likelihood ratio chi-square test and $P < 0.05$ for significance to test associations between left atrial enlargement severity, right atrial enlargement, and valentine shape severity. P -values reported were those for excluding the effects from the full three-way interaction model. Analysis was done using all cats for which measurements were made for all the variables being analyzed.

Results

Eighty-one cats met the inclusion criteria. Five cats were imaged twice during the study time period, however to maintain independent observations and avoid bias, only the earliest images of each (i.e. their initial presentation) were used. Dorsoventral thoracic radiographs were evaluated in 40 cats and VD radiographs were evaluated in 41 cats. Forty-five (56%) cats were diagnosed with HCM, ten (12%) were diagnosed with HOCM, and 26 (32%) were diagnosed with another “miscellaneous” condition (Table 1). This

miscellaneous category included the diagnoses of congenital septal and valvular defects (7), restrictive cardiomyopathy (2), unclassified cardiomyopathy (4), primary arrhythmias (1), hyperthyroidism (5), volume overload (2), and normal (5). These diagnoses were based on echocardiographic findings, T4 values, blood pressure, and the clinicians’ assessment of the patient. These diagnoses were not confirmed with necropsy or selective angiography. Because of the numerous final diagnoses, the diagnoses were grouped

TABLE 1. Clinical Diagnoses Assigned to Each of 81 Cats Evaluated^a

Diagnosis	Number of cats
Hypertrophic cardiomyopathy (HCM)	45
Hypertrophic obstructive cardiomyopathy (HOCM)	10
Total	55
Miscellaneous:	
Congenital septal and valvular defects	7
Restrictive cardiomyopathy	2
Unclassified cardiomyopathy	4
Primary arrhythmias	1
Hyperthyroidism	5
Volume overload	2
Normal	5
Total	26

^aA cardiac diagnosis was assigned to each cat based on echocardiographic findings, blood pressure, T4 values, and other clinical findings. For statistical analysis, HCM and HOCM cats were evaluated independently and together in comparison to the remaining miscellaneous diagnoses.

TABLE 2. Subjective Radiographic Scoring of the Valentine-Shaped Cardiac Silhouette for Included Cats

Radiographic score of valentine shape	DV/VD ^a
1 = mild	16
2 = moderate	39
3 = severe	26

^aDorsoventral (DV) or ventrodorsal (VD) thoracic radiographic projections were evaluated.

two different ways for separate analyses. The first compared HCM ($n = 45$), hypertrophic obstructive cardiomyopathy (HOCM) ($n = 10$), and all other diagnoses categorized as “miscellaneous” ($n = 26$). A second grouping combined HCM and HOCM (55 cats total) and compared these to the remaining “miscellaneous” diagnoses. There was no statistical significance between HCM, HOCM, miscellaneous conditions, and the severity of valentine shape on radiographs ($P = 0.23$). Similarly, there was no statistical significance when cats with HCM and HOCM were collectively evaluated in regards to valentine shape severity ($P = 0.52$).

For the severity of radiographic evidence of valentine-shaped cardiac silhouette, 16 (20%) cats had a radiograph score of 1 (mild), 39 (48%) had a score of 2 (moderate), and 26 (32%) had a score of 3 (severe) (Table 2). Of the 81 echocardiograms reviewed, 61 (75%) had a subjective echocardiographic right atrial enlargement score of 1 (normal right atrium), 20 (25%) had a subjective echocardiographic right atrial enlargement score of 2 (mild/moderate enlargement) and no cat had a score of 3 (severe enlargement). Fifteen cats (18%) had a subjective echocardiographic left atrial enlargement score of 1 (normal left atrium), 20 (25%) had a score of 2 (mild/moderate enlargement) and 46 (57%) had a score of 3 (severe enlargement) (Table 3).

There was a strong positive correlation between severity of left atrial enlargement and the severity of valentine shape ($P < 0.001$) (Table 4). The observed frequency of moderate or severe valentine shape increased from 40% (6/15) to 70% (14/20) to 98% (45/46) with increasing severity of left atrial enlargement from normal to mild/moderate to severe. Similarly the observed frequency of severe valentine shape specifically increased from 7% (1/15) to 15% (3/20) to 48% (22/46) with increasing severity of left atrial enlargement. There was no pairwise association between

TABLE 4. Comparison Between Radiographic Scores of Valentine Shape Severity and Echocardiographic Scores of Left Atrial Enlargement for Included Cats^a

Radiographic score of valentine shape	Echocardiographic score of left atrial enlargement		
	Normal	Mild/moderate	Severe
Mild	9	6	1
Moderate	5	11	23
Severe	1	3	22

^aWhen evaluated for subjective evidence of left atrial enlargement on echocardiogram and severity of valentine shape of cardiac silhouette on thoracic radiographs, there was a strong correlation ($P < 0.001$) between left atrial size and valentine severity.

the degree of echocardiographic right atrial enlargement and the severity of valentine shape on thoracic radiographs ($P = 0.36$). However, there was a significant ($P = 0.04$) three-way interaction, which suggested that the effect of right atrial enlargement on severity of valentine shape was conditional on the degree of left atrial enlargement (Fig. 3). Specifically, when concurrent with severe left atrial enlargement, the presence of right atrial enlargement increased the observed frequency of a severe valentine shape from 41% (15/37) to 78% (7/9). No effect of right atrial enlargement was noted when left atrial enlargement was absent or only mild/moderate with the observed frequency of a severe valentine shape being less (9% (1/11) vs. 13% (3/24)) when right atrial enlargement was present. There was no statistically significant correlation between the presence of left atrial enlargement and the presence of right atrial enlargement on echocardiography ($P = 0.21$).

Fifty (62%) cats were scored as having radiographic evidence of left atrial enlargement on lateral radiographs alone. Although not the primary objective of this study, there was a significant correlation between the subjective degree of left atrial enlargement based on echocardiography and the consensus score of the presence of absence of left atrial enlargement based on the lateral thoracic radiographs ($P < 0.001$).

Discussion

Results of the current study supported the hypothesis that a valentine-shaped cardiac silhouette in feline dorsoventral or ventrodorsal radiographs is primarily due to left atrial enlargement. There was a significant correlation between left atrial echocardiographic score and the valentine shape severity on radiographs. Significant increases in the frequency of moderate and/or severe valentine shape were seen with increasing left atrial size, and right atrial enlargement affected the severity of a valentine-shaped cardiac silhouette only when severe left atrial enlargement was also present. There was no statistical

TABLE 3. Subjective Echocardiographic Scoring of Left and Right Atrial Enlargement for Included Cats^a

Echocardiographic score of atrial enlargement	Left atrium	Right atrium
1 = normal, no enlargement	15	61
2 = mild/moderate enlargement	20	20
3 = severe enlargement	46	0

^aM-mode and 2D echocardiogram images were evaluated to assign scores.

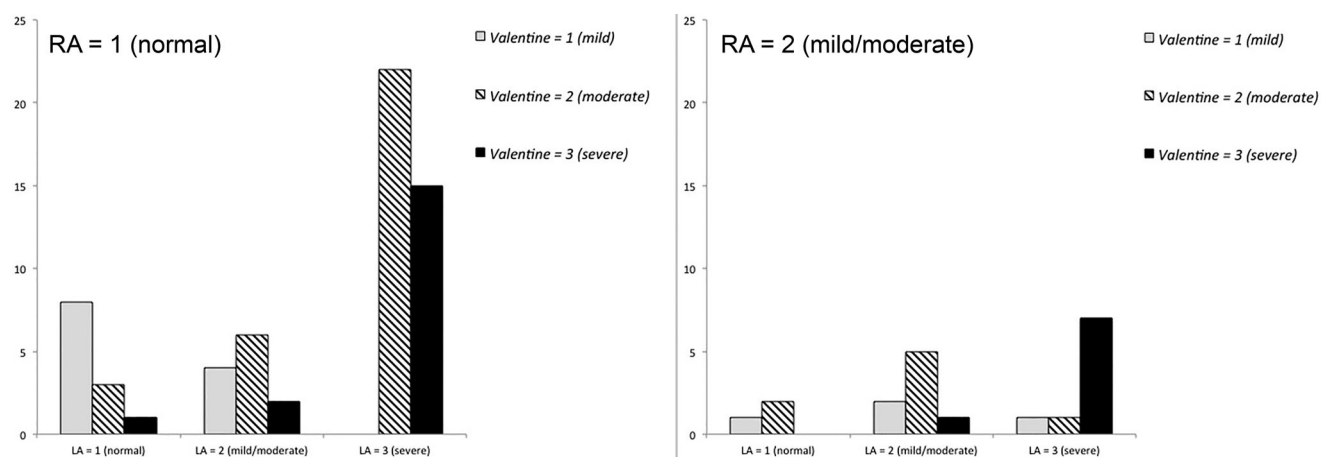


FIG. 3. Three-way relationship between cats with echocardiographic evidence of left and/or right atrial enlargement and radiographic evidence of a valentine-shaped cardiac silhouette. There is a clear association between the size of the left atrium (LA = 1-normal; 2-mild/moderate enlargement; 3-severe enlargement) and the extent of the valentine appearance of the cardiac silhouette (Valentine = 1-mild; 2-moderate; 3-severe) whether the right atrium is normal (RA = 1) or has mild/moderate enlargement (RA = 2). No effect of mild/moderate right atrial enlargement (RA = 2) on valentine appearance is noted unless left atrial enlargement is also present. No cats had severe right atrial enlargement.

significance between the diagnosis (e.g. HCM, HOCM, etc.) and the severity of the valentine shape on radiographs.

This study differs from previous reports of the valentine shape of feline cardiomegaly in that the number of echocardiograms and thoracic radiographs reviewed was higher (81 vs. 35) and echocardiographic images were rereviewed to obtain a qualitative judgment of right atrial size.³ Our conclusions were similar to those of previous studies in that most cats with a valentine-shaped cardiac silhouette had left atrial rather than biatrial enlargement.

One limitation of the current study was the reliance on subjective scoring, though using consensus scoring on radiographs likely minimized interobserver variability. Although there are various methods of quantifying left atrial size in feline cardiomyopathy via echocardiography,⁶ there is no consensus for quantification of right atrial size and, thus, clinicians rely on subjective echocardiographic assessment of the size of the right atrium. Therefore, in this study, qualitative echocardiographic assessments of left and right atrial enlargement were used in addition to qualitative assessment of the degree of severity of the valentine shape of the cardiac silhouette. We used both VD and DV radiographs when evaluating the severity of the valentine shape. Though DV radiographs are often used for cardiac

evaluations, differences between the two projections are minimal in cats, though the appearance of the cardiac silhouette may be more consistent on DV projections.⁷ Another limitation of the study was that cardiac silhouette appearance was not compared to signalment data such as gender, age, breed, and body condition. Specific data on signalment and body condition were not collected for inclusion in the analysis as the association between radiographic and echocardiographic features was our primary concern and we were interested in evaluating this association across a global hospital population. The study was also limited by a lack of angiographic or postmortem confirmation of the clinical diagnosis.

In conclusion, most cats with a valentine-shaped cardiac silhouette on thoracic radiographs in this sample population had left atrial rather than biatrial enlargement. This information can be used to assist radiographic interpreters in narrowing the list of differential diagnoses by ruling out anomalies expected to lead to primarily right atrial enlargement, such as cor triatriatum dexter and isolated tricuspid insufficiency. On the other hand, as there was no relationship between the severity of the valentine shape and the specific type of heart disease in the current study, and authors recommend that echocardiography still be done to confirm the diagnosis.

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